

APTITUDE (QUANT) REASONING QUICK CHEATSHEET

LAST-MINUTE PLACEMENT EXAM REVISION (1-2 MIN SCAN)



NUMBER SYSTEM & AVERAGES Σ

- Sum of N numbers: $\frac{N(N+1)}{2}$
- Average: Sum / N
- Average of first N odd/even numbers



Sum: 15
Average: 3



PERCENTAGE & FRACTIONS $\%$

- Percentage of value: $(\text{Part}/\text{Whole}) \times 100$
- Percentage Change: $\left(\frac{\text{New}-\text{Old}}{\text{Old}}\right) \times 100$
- Multiplication factors: $25\% = 1/4$
 $50\% = 1/2$



PROFIT & LOSS

- Profit: $SP - CP$ Loss: $CP - SP$
- Profit%: $\frac{P}{CP} \times 100$ Loss%: $\frac{L}{CP} \times 100$
- Effective discount formula:
 $50\% \text{ off} + 50\% \text{ off} \neq 100\% \text{ off}$



TIME & WORK

- Work = Rate * Time
- Total work: Days for A & B = $\frac{A \times B}{A+B}$
- Tank filling rates: A A_1 & B = A & C C_1
- Efficiency: 



DATA INTERPRETATION (DI)

- Analyzing charts and tables
- Bar chart
- Examples: 
- Growth rate: $= \frac{\text{Growth rate}}{100\%}$



SPEED, TIME & DISTANCE

- Distance = Speed * Time
- Average Speed: $\frac{2xy}{x+y}$
- Upstream/Downstream speed for boats: $\Rightarrow \Rightarrow$ boats
- Relative speed for trains: 



SIMPLE & COMPOUND INTEREST

- SI: $\frac{(P \times R \times T)}{100}$
- CI: $P \left(1 + \frac{R}{100}\right)^T - P$
- CI amount: $P \left(1 + \frac{R}{100}\right)^T$
- SI: $\frac{P \times R \times T}{100}$
- CI: $P \left(1 + \frac{R}{100}\right)^T - P$
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PERMUTATION & COMBINATION (P&C)

- $nPr: \frac{n!}{n! / (n-r)!}$
- $nCr: \frac{n!}{r! * (n-r)!}$
- Fundamental principle of counting (product (product rule))
- $nCr: \frac{n!}{r! * (n-r)!}$
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- Profit%: $\frac{P}{CP} * 100$ • Loss%: $\frac{L}{CP} * 100$
- Effective discount formula:
 $SP = \% / CP + CP - SP = \% CP$



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- Efficiency:



BETICK & NICIT



- Context tent: $(N + N)$
- Average of riscount formula:
 $V = \frac{\text{Profit}}{N + N} \sim x \frac{\text{Sum}}{(n - N)}$
- Effective discount formula:
 $20\% = \frac{(CP - CP)}{AB}$



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